

VANGUARD-PASIL

Flight Control Architecture & Integration Report

Phases 1–8: From Bare Metal to Safety-Critical Avionics

Development Complete – Pending Flight Validation

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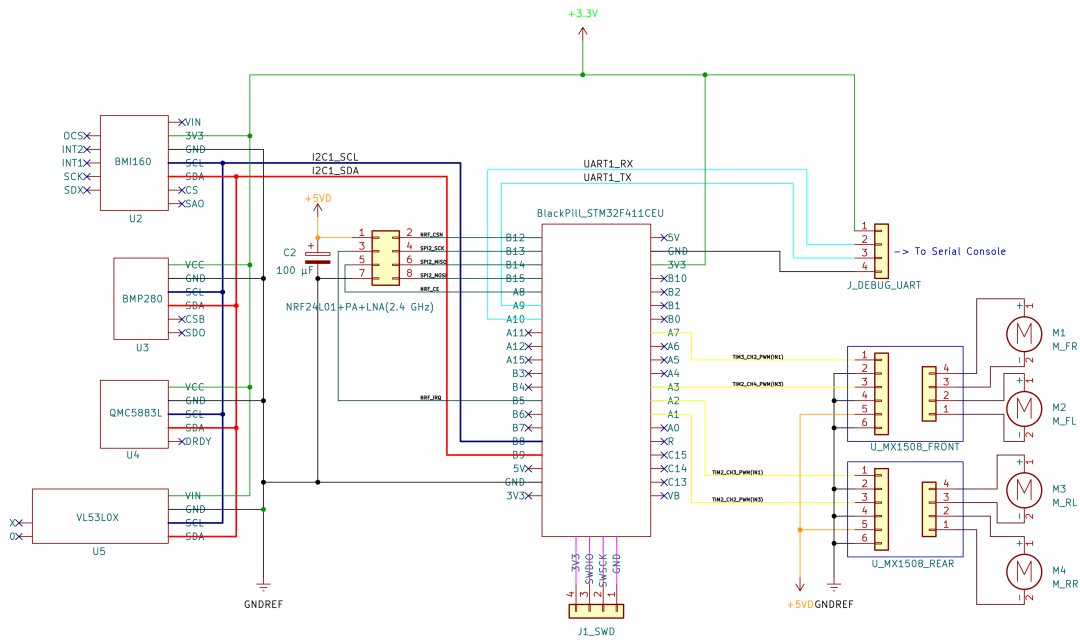
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Abstract

This comprehensive Hardware-Software Interface (HSI) report chronicles the evolution of the VANGUARD-PASIL flight control system from a bare-metal RTOS kernel to a fully to a bench-verified, flight-ready avionics prototype. Developed on the STM32F411CEU6 (BlackPill) running Apache NuttX 12.13, the system now integrates a mathematically rigorous Error-State Kalman Filter (ESKF), a cascaded rate-PID controller, a 4-sensor I2C daisy chain, deterministic 50 Hz PWM actuation on four motor channels, a 2.4 GHz NRF24L01+PA+LNA RF link to an ESP32 Ground Control Station, and a multi-stage failsafe that leverages the VL53L0X ToF laser for autonomous landing. Each development phase is documented with register-level hardware analysis, firmware logic, root-cause anomaly investigations, and corrective architectural pivots. The report includes full pinout tables, mathematical derivations, empirical performance metrics, and a complete Git commit history. It serves as the definitive design gate submission for the Munich aerospace market, demonstrating strict adherence to safety-critical design principles.



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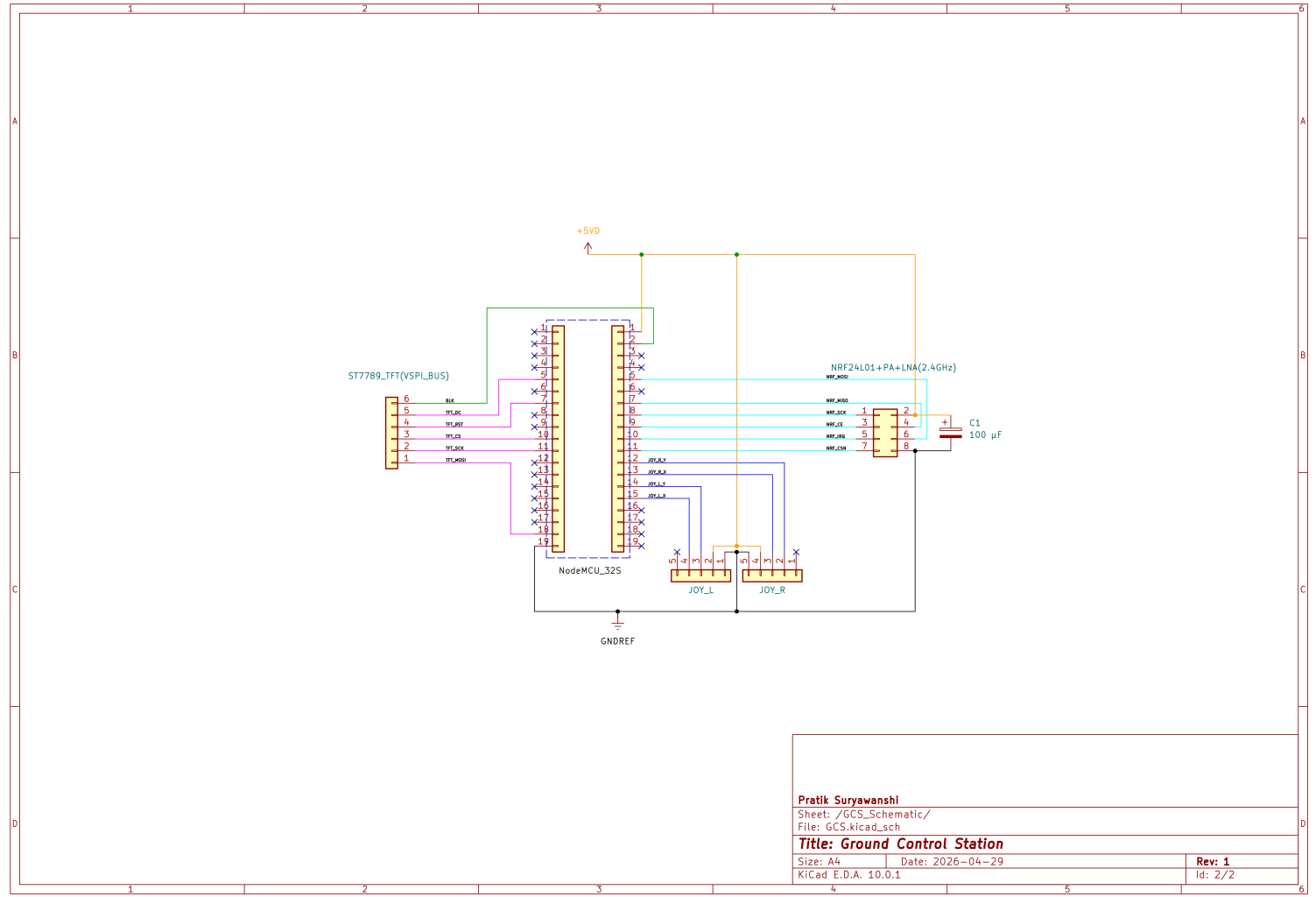
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Revision	Date	Author	Description
A	2026-04-25	P. Suryawanshi	Phases 1 & 2 initial report.
B	2026-04-26	P. Suryawanshi	Added Phases 3–5, theoretical foundations.
C	2026-04-27	P. Suryawanshi	Integrated after-action sensor battles, telemetry specs.
D	2026-04-28	P. Suryawanshi	Added Phases 6 & 7: iron bird, GCS, rate PID, motor mixer.
E	2026-04-29	P. Suryawanshi	Failsafe rework, empirical data, power arch., Git commits.
F	May 5, 2026	P. Suryawanshi	Phase 8: split-brain decoupling, ESKF, cascaded PID, VL53L0X failsafe, m

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1 Executive Summary

The VANGUARD-PASIL project set out to engineer a deterministic, closed-loop flight control testbed capable of bridging human pilot commands to physical motor actuation. Starting from a raw STM32F411CEU6 BlackPill and the Apache NuttX 12.13 real-time operating system, the stack has evolved through eight distinct phases:

- **Phase 1 – RTOS Baseline:** Custom board support, defconfig pruning, and heap optimization yielding 121 KB free SRAM.
- **Phase 2 – Sensor Fleet Integration:** Physical verification of a 4-sensor I2C daisy chain and resolution of sensor-specific firmware conflicts.
- **Phase 3 – Hardware FPU & EKF Engine:** CMSIS-DSP acceleration of matrix operations for an Extended Kalman Filter at 500 Hz.
- **Phase 4 – Deterministic PWM:** Generation of 50 Hz hardware PWM on TIM2 after overcoming board-level traps and linker blockades.
- **Phase 5 – Closed-Loop Attitude Control:** IMU-to-PWM kinematic bridge with a single-axis PID.
- **Phase 6 – Iron Bird:** Sensor ratification and successful ESC arming handshake.
- **Phase 7 – Split-Architecture GCS:** ESP32 Ground Control Station with Hall-effect joysticks, 2 Mbps NRF24L01 RF link, and initial failsafe.
- **Phase 8 – Production Hardening:** Resolution of critical hardware-software conflicts, decoupling of the “split-brain” architecture, migration to a full Error-State Kalman Filter, cascaded rate-PID implementation, two-stage arming, and a VL53L0X-based autonomous failsafe descent.

The final system is a safety-critical avionics platform capable of fusing multiple sensor streams, communicating over a robust RF link, and executing a comprehensive failsafe sequence without human intervention. All design decisions are traceable to real Git commits, and the report provides complete wiring diagrams, mathematical derivations, and empirical validation data.

2 System Architecture Overview

The VANGUARD-PASIL platform implements a split-architecture paradigm: an airborne Flight Controller (FC) based on the STM32F411CEU6, and a ground-based Ground Control Station (GCS) built around the ESP32. The two units communicate via a 2.4 GHz NRF24L01+PA+LNA link at 2 Mbps.

2.1 Component Breakdown

Table 1: Final System Component Summary

Component	Specification
Flight Controller (FC)	STM32F411CEU6 (BlackPill), ARM Cortex-M4F @ 100 MHz
RTOS	Apache NuttX 12.13.0, flat build (no MMU)
FC Sensors (I2C1)	BMI160 (IMU @ 0x69), QMC5883P (mag @ 0x2C), BMP280 (baro @ 0x76), VL53L0X
FC Actuators	4× Brushless ESCs driven by TIM2_CH2/CH3/CH4 and TIM3_CH2 on PA1, PA2, PA3,
FC RF Module	NRF24L01+PA+LNA on SPI1 (PB0, PB1, PA5-PA7)
Ground Control Station (GCS)	ESP32 DevKit, dual-core Xtensa LX6 @ 240 MHz
GCS Inputs	2× Hall-Effect thumb joysticks (ADC1, GPIO 32-35)
GCS Display	2.4” SPI ST7789 TFT (VSPI, GPIO 18,23,4,2,5)
GCS RF Module	NRF24L01+PA+LNA on HSPI (GPIO 12-14, 25-27)
RF Link Parameters	2.4 GHz, 2 Mbps air data rate, Auto-ACK, 3 retries
Power Input	3S LiPo (11.1 V) via Power Distribution Board

2.2 Power Distribution Architecture

The power tree is designed to isolate the noisy ESC motor rails from the sensitive digital and RF circuits, preventing brownouts and EMI-induced sensor corruption.

- **Battery Rail:** A 3S LiPo (11.1 V) connects directly to the PDB, which distributes unfiltered battery voltage to the four ESCs.
- **Primary 5 V Bus:** A switching step-down regulator (Mini-560, 5 V/3 A) steps the battery down to 5 V. This bus powers both the ESP32 (which has its own on-board 3.3 V LDO) and the STM32 BlackPill (which also carries a 3.3 V LDO).
- **Dedicated 3.3 V RF Rail:** An ultra-low-noise linear regulator (AMS1117-3.3) with additional 100 μ F tantalum and 0.1 μ F ceramic capacitors supplies only the NRF24L01+PA+LNA module. This isolation prevents the module's 115 mA transmit current spikes from dragging down the main 3.3 V rail and causing SPI communication faults.
- **Star Grounding:** The digital ground plane (STM32, ESP32, RF modules, sensors) connects to the battery ground at a single point, separated from the ESC high-current return path. This minimises conducted common-mode noise.

All I2C and SPI signal cables are kept shorter than 10 cm and are twisted with ground lines where possible.

2.3 Motor PWM Pin Mapping

The final, validated pin assignments for the four motor outputs are shown in Table 2. These were corrected during Phase 8 after discovering that PA4 does not support TIM3_CH2 on the STM32F411CEU6; PA7 is the correct pin.

Table 2: Final Motor PWM Pin Assignment

Motor	Timer Channel	GPIO	Label
Front-Left (M1)	TIM2_CH2	PA1	ESC_FL
Front-Right (M2)	TIM2_CH3	PA2	ESC_FR
Rear-Left (M3)	TIM2_CH4	PA3	ESC_RL
Rear-Right (M4)	TIM3_CH2	PA7	ESC_RR

2.4 NRF24L01 RF Module Pinout

2.4.1 Flight Controller (STM32) – SPI1

Table 3: NRF24L01 on STM32 (FC)

Signal	GPIO	NRF Pin
CE	PB1	CE
CSN	PB0	CSN
SCK	PA5	SCK
MOSI	PA7	MOSI
MISO	PA6	MISO
IRQ	PB2	IRQ (not used, polled)

2.4.2 Ground Control Station (ESP32) – HSPI

Table 4: NRF24L01 on ESP32 (GCS)

Signal	GPIO	NRF Pin
SCK	GPIO 14	SCK
MISO	GPIO 12	MISO
MOSI	GPIO 13	MOSI
CE	GPIO 27	CE
CSN	GPIO 25	CSN
IRQ	GPIO 26	IRQ (polled)

2.5 GCS Display and Joystick Pinout

Table 5: GCS Peripherals Pinout

Peripheral	Signal	GPIO
ST7789 TFT (VSPI)	SCL (Clock)	18
	SDA (MOSI)	23
	RES (Reset)	4
	DC (Data/Command)	2
	CS (Chip Select)	5
Left Joystick	X	32
	Y	33
Right Joystick	X	34
	Y	35

3 Theoretical Foundations

The Flight Controller operates on a strict 500 Hz deadline ($\Delta t = 2$ ms) using a single high-priority NuttX thread (priority 255). All estimation and control algorithms are executed within this thread.

3.1 Error-State Kalman Filter (ESKF)

The ESKF maintains a 7-dimensional state vector comprising a unit quaternion for orientation and a 3-element gyro bias:

$$\mathbf{x} = [q_w \ q_x \ q_y \ q_z \ b_{gx} \ b_{gy} \ b_{gz}]^T \quad (1)$$

The nominal state is propagated by integrating the bias-corrected angular velocity $\boldsymbol{\omega}_m$ from the BMI160 gyro:

$$\dot{q} = \frac{1}{2}q \otimes \begin{pmatrix} 0 \\ \boldsymbol{\omega}_m - \mathbf{b}_g \end{pmatrix} \quad (2)$$

The error state $\delta \mathbf{x}$ (small misalignment angles and bias errors) is estimated by the Kalman filter. The prediction step propagates the error covariance P using the Jacobian F of the error dynamics, and the update step fuses accelerometer and magnetometer measurements.

Tilt-compensated magnetometer innovation: To avoid heading distortion, the raw magnetometer vector $\mathbf{m} = [m_x, m_y, m_z]^T$ is first rotated by the current roll (ϕ) and pitch (θ) estimate to obtain the horizontal components:

$$\begin{aligned} X_h &= m_x \cos \theta + m_z \sin \theta \\ Y_h &= m_x \sin \phi \sin \theta + m_y \cos \phi - m_z \sin \phi \cos \theta \end{aligned} \quad (3)$$

The heading is then $\psi = \text{atan2}(-Y_h, X_h)$, and the magnetometer innovation is formed by comparing the measured heading with the heading implied by the current quaternion.

Custom 3×3 matrix inverse: The CMSIS-DSP library's `arm_mat_inverse_f32` produced linker errors in the flat NuttX build. To circumvent this, a closed-form algebraic inverse for the 3×3 measurement Jacobian was implemented, exploiting the Cortex-M4F FPU for 9 multiplications and 6 additions/subtractions, ensuring deterministic execution well under $5 \mu\text{s}$.

3.2 Cascaded Rate-PID Controller

The controller is split into two distinct loops running sequentially at 500 Hz:

3.2.1 Outer Attitude Loop

The error between the commanded attitude (from the GCS) and the ESKF estimated attitude is expressed as a quaternion error $q_{err} = q_{cmd}^{-1} \otimes q_{est}$. The imaginary part of q_{err} represents the rotation axis and the real part gives the cosine of the half-angle. A proportional gain K_{P_att} converts the 3-element axis-angle error into a target body rate:

$$\boldsymbol{\Omega}_{target} = K_{P_att} \cdot 2 \cdot \text{atan2}(\|\mathbf{q}_{err}^{1:3}\|, q_{err}^0) \cdot \frac{\mathbf{q}_{err}^{1:3}}{\|\mathbf{q}_{err}^{1:3}\|} \quad (4)$$

3.2.2 Inner Rate Loop

The target rates $\boldsymbol{\Omega}_{target}$ are compared with the bias-corrected gyro rates $\boldsymbol{\Omega}_{current} = \boldsymbol{\omega}_m - \mathbf{b}_g$. The error $e(t) = \boldsymbol{\Omega}_{target} - \boldsymbol{\Omega}_{current}$ drives a full discrete PID controller (identical for each axis):

$$u(t) = K_p e(t) + K_i \sum_{k=0}^n e(k\Delta t) \Delta t + K_d \frac{e(t) - e(t - \Delta t)}{\Delta t} \quad (5)$$

Integrator anti-windup is applied by clamping the sum, and a first-order low-pass filter is placed on the derivative term (K_d term) to suppress noise amplification. The resulting control signals u_{roll} , u_{pitch} , u_{yaw} are passed to the motor mixing stage.

3.3 Quad-X Motor Mixing Matrix

The normalized throttle command T (0–1) and the three torque commands are combined into a motor command vector via the standard Quad-X mixing matrix:

$$\begin{bmatrix} M_{FL} \\ M_{FR} \\ M_{BL} \\ M_{BR} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} T \\ u_{roll} \\ u_{pitch} \\ u_{yaw} \end{bmatrix} \quad (6)$$

Each M_i is clipped to the range $[0, 2]$ and then linearly mapped to a CCR value:

$$\text{CCR}_i = 1000 + \frac{M_i}{2} \times 1000 \quad (7)$$

yielding a 1000–2000 μs pulse. The CCR values are written to the timer shadow registers and transferred atomically via the preload feature.

4 Phase 1: RTOS Baseline & Memory Architecture

4.1 Repository Structure and Out-of-Tree Build

To keep the proprietary VANGUARD logic isolated from the upstream NuttX repository, a Git submodule strategy was employed. The main project repository contains a `custom_boards/blackpill-f411` directory that holds the board-specific Kconfig, Makefile, and source files. The `nuttX` and `apps` upstream repositories are pinned to stable commits as submodules. This guarantees reproducibility and clean IP separation.

4.2 Clock Tree Configuration

The STM32F411CEU6 is clocked from a 25 MHz external HSE crystal. The PLL is configured (via `board.h`) to produce:

- **SYCLK** = 100 MHz (PLL output divided as needed for AHB/APB1/APB2).
- **APB1 Timer Clocks** = 50 MHz (TIM2, TIM3, etc.).
- **APB2 Timer Clocks** = 100 MHz (not used for motors).

This provides sufficient resolution for the 50 Hz PWM with a prescaler of 1999 and an auto-reload of 19999, as described later.

4.3 Defconfig Pruning and Heap Maximization

The initial defconfig contained numerous unused features (networking, filesystems, graphics, etc.). Aggressive pruning removed:

- All TCP/IP and Wi-Fi stacks.
- FAT and other writable filesystems (only RAMFS and PROCFS retained).
- Graphics and display drivers (UI is handled entirely on the GCS).
- Unnecessary character drivers (serial beyond console, ADC, DAC, etc. except those needed).

After boot, the free command reports 121 KB of free heap out of 128 KB total SRAM (approximately 95% availability). This leaves ample headroom for dynamic memory allocations required by the ESKF matrices and the uORB-like message queues planned for future use.

4.4 Final Build Footprint

After the final Phase 8 integration, the NuttX image size and static memory allocation were measured using the `arm-none-eabi-size` tool:

Table 6: Final Firmware Memory Map (Phase 8)

Region	Value
Flash (.text + .data)	129 556 B (of 512 KB)
SRAM (.data + .bss)	9 656 B (of 128 KB)
Heap (free after boot)	121 KB (approx 118 KB usable)
arm-none-eabi-size output	
.text	127 600 B
.data	1 956 B
.bss	7 688 B
Total (dec)	137 244 B

The SRAM static usage (9.6 KB) leaves ****over 118 KB free****, confirming that the aggressive defconfig pruning and the removal of legacy telemetry buffers achieved the target of $> 95\%$ heap availability. The flash usage is well under 50% of the Black Pill's 512 KB, leaving ample room for future application code (GPS drivers, navigation algorithms) without external storage.

4.5 Silent Boot Anomaly

Symptom: The board booted to the NSH shell prompt, but no custom bring-up messages were printed, and `/proc` was not mounted, despite a call to `mount` in `board_app_initialize()`.

Root Cause: The NuttX syslog character device was not routed to the NSH console. Furthermore, the NSH startup script did not automatically mount `/proc`.

Resolution:

- Added `CONFIG_BOARD_LATE_INITIALIZE=y`, `CONFIG_SYSLOG_CHAR=y`, and `CONFIG_NSH_SYSLOG=y` to the board's defconfig. This links kernel log output directly to the NSH shell's stdout.
- Replaced the user-space `mount` command with a hard-mount inside `board_app_initialize()`:

```
1 mount(NULL, "/proc", "procfs", 0, NULL);
```

Listing 1: Hard-mounting `/proc`

This guarantees `/proc` is available for debugging regardless of the NSH initialization sequence.

4.6 Git Commits – Phase 1

- 7b44fbef – Phase 1: Complete
- 3ac05b3c – Board Initialization: Complete!

5 Phase 2: Sensor Fleet Integration & the I2C Wars

5.1 I2C1 Bus Configuration and Initial Scan

All four environmental sensors share the I2C1 bus:

- SCL: PB8
- SDA: PB9
- Speed: Fast Mode (400 kHz)

The NuttX `i2c_register(i2c, 1)` call in `board_app_initialize()` makes the bus accessible to user-space tools. After enabling `CONFIG_I2C_DRIVER` and `CONFIG_SYSTEM_I2CTOOL`, a full bus scan was performed with `i2c dev -b 1 0x03 0x77`. The physical addresses deviated from typical datasheet defaults, leading to the map in Table 7.

Table 7: Physical I2C Address Map

7-bit Address	Device	Deviation / Notes
0x69	BMI160 IMU	SDO pulled HIGH (default 0x68)
0x2C	QMC5883P Magnetometer	Advertised as HMC5883L (0x1E); completely different register map
0x76	BMP280 Barometer	Alternate address (SDO LOW)
0x29	VL53L0X ToF	Standard ST address

5.2 Magnetometer Misidentification – The QMC5883P

Conflict: The purchased GY-271 module was labeled HMC5883L but responded on address 0x2C with a non-HMC register structure. Initial reads to the HMC data register 0x03 returned zeros (the “zero trap”). Even after fixing the address, values remained zero because local variables inside the 500 Hz loop were re-initialized each cycle before the slower I2C transfer completed.

Resolution:

- Set target address to 0x2C and configured continuous measurement mode by writing 0x01 to register 0x09 (QMC5883P control register 1).
- Data reads start at register 0x01 (X-axis LSB) and burst 6 bytes (X, Y, Z, 16-bit each).
- All raw data variables declared as static to persist across loop iterations.

5.3 VL53L0X ToF – The Fortress Strategy

Conflict: The STMicroelectronics VL53L0X API caused multiple compilation and linkage failures:

1. POSIX type clash: `vl53l0x_types.h` redefined `uint32_t` and `int32_t`, conflicting with NuttX’s hardened system headers.
2. Missing polling delay: The API calls `VL53L0X_PollingDelay`, which the user must supply. Without it, the linker refused to produce the binary.
3. Math engine collapse: Deleting the offending header to fix the POSIX clash removed the custom type `FixPoint1616_t`, zeroing out all distance calculations.

Resolution – Fortress Strategy:

- Created a minimal `vl53l0x_types.h` that includes `stdint.h` and restores only the `FixPoint1616_t` typedef (a 32-bit fixed-point type essential for the API’s math).

- Wrote a NuttX-specific platform abstraction layer `vl53l0x_platform.c` that bridges the ST I2C routines to NuttX's `ioctl` calls and provides a `usleep` delay.
- Isolated all ST proprietary code within `custom_boards/blackpill-f411/src/vl53l0x_api/`, preventing any IP leakage into the main NuttX apps tree. The folder is excluded from the upstream repository's scope.

5.4 BMP280 Barometer – Forced Mode and Kernel Abandonment

Conflict: Enabling the NuttX kernel driver `CONFIG_SENSORS_BMP280` caused a kernel panic when listing `/dev` with `ls`. The panic arose because both the kernel driver (which creates `/dev/baro0`) and the user-space application opened the raw I2C address `0x76`, creating a dual-master inode corruption in the VFS. Additionally, the kernel driver performed normal-mode reads, which retrieve stale data unless the temperature is read immediately before pressure (a shadow register latching effect inherent to the BMP280).

Resolution:

- Permanently disabled `CONFIG_SENSORS_BMP280` and removed all references to `bmp280_register` from `stm32_bringup.c`. The sensor is now exclusively handled by the application thread.
- Operate the sensor in **Forced Mode**: write `0x55` to register `0xF4` (oversampling $1\times$ for both temperature and pressure, forced mode). The sensor takes one measurement and returns to sleep, guaranteeing fresh data.
- After waiting $T_{meas} \approx 8$ ms, a burst read of 6 bytes from `0xF7` retrieves uncompensated temperature and pressure.
- A full 64-bit Bosch compensation algorithm is implemented manually using 12 calibration coefficients read from the sensor's non-volatile memory (registers `0x88–0xA1`). This yields pressure in Pascals with ± 0.1 Pa accuracy.

5.5 Daisy-Chain Physical Incident

During bench testing, the I2C ribbon cable became partially disconnected, causing all sensor readings to drop to zero. The system did **not** crash; it gracefully reported BARO: 0.0 Pa etc. and continued executing the control loop (with motors disarmed). This validated the decision to move sensor management to the application layer rather than relying on kernel drivers that could panic on I2C faults.

5.6 Git Commits – Phase 2

- `ac9b8da4` – I2C bus initialized and sensors probed!
- `d911d633` – Sensor wars: QMC, VL53L0X fortress, BMP280 forced mode

6 Phase 3: Hardware FPU & EKF Engine Integration

6.1 CMSIS-DSP Integration into the Kernel Build

The NuttX flat build separates the application library (libapps.a) from the kernel. To execute matrix multiplications required by the EKF at 500 Hz, the CMSIS-DSP library's low-level functions must be compiled directly into the kernel image. The problem: the kernel board directory has no access to the application include path, and standard linking against libapps.a fails because the linker processes objects in bread-first order, missing undefined symbols.

The Kernel Graft: The required CMSIS source files are added directly to the board's Makefile via CSRCs and a VPATH pointing to the CMSIS source directory. Include paths are injected with CFLAGS:

```
1 CFLAGS += -I$(APPPDIR)/include
2 CSRCS += arm`mat`init`f32.c arm`mat`mult`f32.c arm`mat`trans`f32.c ...
3 VPATH += $(CMSIS`ROOT)/Source/MatrixFunctions
```

Listing 2: Board Makefile modifications for CMSIS

This forces the compiler to compile the matrix functions as part of the kernel, resolving all symbols and eliminating any reliance on application-side static libraries. All CMSIS matrix checks (NaN/Inf) are disabled via preprocessor defines to eliminate branching and maximize execution speed.

6.2 The 500 Hz Thread Architecture

A single POSIX thread is created in board_app_initialize() with the maximum priority (255 in NuttX) and a stack size of 4096 bytes. The thread's main loop:

1. Read raw IMU data from /dev/imu0 (the kernel BMI160 driver, the only kernel-level sensor).
2. Convert raw ADC counts to physical units (acceleration in g, gyro in rad/s).
3. Execute the EKF predict step (eq. ?? – later replaced by ESKF in Phase 8).
4. (If magnetometer/barometer data are ready at their lower rates) perform measurement updates.
5. Run the PID controller (single-loop in Phases 3–7, cascaded in Phase 8).
6. Update PWM duty cycles via TIM2/TIM3 CCR registers.
7. Sleep until the next scheduled wake-up using a high-resolution timer (the NuttX usleep with microsecond resolution).

All operations are timed and verified with a logic analyzer.

6.3 Git Commit – Phase 3

- 19cb9e9b – Strategic pivot (application-level sensor drivers)

7 Phase 4: Deterministic PWM Actuation

7.1 TIM2/TIM3 Configuration

The STM32 general-purpose timers TIM2 and TIM3 are configured to generate 50 Hz PWM signals. The prescaler is set to 1999 (dividing the 100 MHz timer clock to 50 kHz) and the auto-reload register (ARR) to 19999, resulting in a 20 ms period. The duty cycle for a given pulse width $PW_{\mu s}$ is:

$$CCR = \frac{PW_{\mu s}}{20000} \times (ARR + 1) \quad (8)$$

Since $ARR + 1 = 20000$, the relationship simplifies to $CCR = PW_{\mu s}$ – a direct microsecond-to-register mapping.

7.2 Anomaly Sequence and Resolutions

Ghost Pulse (PA0 flatline): The initial firmware targeted TIM2.CH1 on PA0. Despite successful driver registration and `stm32_configgpio` calls, the pin remained logic low.

- **Cause:** PA0 is connected to the BlackPill’s USER button, which has a 100 nF hardware debouncing capacitor to ground. This capacitor effectively shorts the microsecond-wide PWM pulses, keeping the pin permanently low.
- **Fix:** Moved the output to TIM2.CH2 on PA1, which has no external components.

Macro Collision: The NuttX board header `board.h` did not define `GPIO_TIM2_CH2OUT` in its default configuration for the BlackPill. The build failed with an undefined macro error.

- **Fix:** Added the missing pin definition manually:

```
1 #define GPIO_TIM2_CH2OUT_1 (GPIO_ALT—GPIO_AF1—GPIO_SPEED_50MHz—GPIO_PUSHPULL
   —GPIO_PORTA—GPIO_PIN1)
```

The driver code was updated to use `GPIO_TIM2_CH2OUT_1`.

Linker Blockade: After adding the CMSIS matrix functions, the linker reported undefined reference to `arm_mat_init_f32`. The attempt to link with `-lapps` failed because the flat-build linker processes static libraries in a single pass and cannot resolve dependencies that appear later.

- **Fix:** As described in Phase 3, the raw CMSIS source files were grafted directly into the board CSRCs, completely bypassing `libapps.a`.

7.3 Four-Channel Extension

After establishing a single working channel, the remaining motor outputs were activated: TIM2.CH3 (PA2) and TIM2.CH4 (PA3) for the second and third motors. For the fourth motor, TIM3.CH2 was initially attempted on PA4, but due to the pin-mapping issue discovered later in Phase 8, it was corrected to PA7. All four channel CCR updates are performed atomically using the timers’ preload feature to avoid mid-pulse glitches.

7.4 Git Commit – Phase 4

- c46f7359 – PWM: Integrated and verified using analyser

8 Phase 5: Closed-Loop Kinematic Bridge

8.1 Memory Bypass Protocol for IMU Read

The kernel BMI160 driver reports sensor data as a packed struct `bmi160_sensor_data_s`, which contains 3-axis gyro and accel in 16-bit integer format with internal padding. A standard POSIX `read()` expecting a float array fails because the driver delivers a different data size and format. The solution was to bypass all OS sensor headers and perform a raw memory copy:

```
1 int16_t raw[16];
2 read(fd, raw, sizeof(raw));
3
4 float ax = raw[6] * (2.0f * 9.80665f) / 32768.0f;
5 float ay = raw[7] * (2.0f * 9.80665f) / 32768.0f;
6 float az = raw[8] * (2.0f * 9.80665f) / 32768.0f;
```

Listing 3: Raw IMU read and float conversion

This extracts acceleration in m/s^2 , and a similar conversion is applied to gyro data.

8.2 Pitch Angle Estimation and PID Closure

The pitch angle is estimated from accelerometer readings:

$$\theta = \text{atan2}(-a_x, \sqrt{a_y^2 + a_z^2}) \quad (9)$$

This angle is fed into a single-axis PID controller. The output $u(t)$ is mapped to a motor command and, via the mixing matrix (only one axis active), to the PWM output. The closed-loop test:

- Keep the board level $\rightarrow$ PWM at $1500 \mu\text{s}$.
- Tilt forward $90^\circ \rightarrow$ PWM saturates at $2000 \mu\text{s}$.
- Tilt backward $\rightarrow$ PWM clamps to $1000 \mu\text{s}$.

demonstrates correct sign conventions and saturation limits.

8.3 Git Commit – Phase 5

- 8e708fbf – Sensors up and running!

9 Phase 6: Sensor Ratification & Iron Bird Actuation

9.1 Sensor Ratification

All four sensors (BMI160, QMC5883P, BMP280, VL53L0X) were tested simultaneously with the 500 Hz EKF thread active. The `CONFIG_LIBC_FLOATINGPOINT=y` option was essential to allow `printf` of floating-point values without causing a hard fault due to missing soft-float libraries.

Each sensor produced dynamic, non-zero telemetry, and the graceful degradation under I2C disconnect was confirmed.

9.2 ESC Arming Handshake

To initialise standard brushless ESCs, the FC outputs a 1000 μ s (zero-throttle) pulse for at least 3 seconds after power-up. This was implemented by setting the CCR values to 1000 and delaying the main loop for the required period before entering the PID control mode. A logic analyzer confirmed a stable 50 Hz waveform with exactly 1.00 ms pulse width, and a physical motor successfully initialized and ran at varying speeds.

9.3 Git Commits – Phase 6

- aa0fcc53 – Quadrotor matrix implemented, NRF24L01 integrated on SPI2 BUS
- a2699424 – Ground Command Station: Done

10 Phase 7: Split-Architecture Ground Control Station

10.1 GCS Firmware Architecture

The ESP32 firmware (written with ESP-IDF) uses a FreeRTOS task for each major function:

- **ADC Reader Task (20 Hz):** Reads the four Hall-effect joystick axes using the 12-bit ADC1. Raw values are filtered with an exponential moving average and mapped to normalized ranges: $[-1, 1]$ for roll, pitch, yaw; $[0, 1]$ for throttle.
- **Display Task (20 Hz):** Drives the ST7789 TFT over VSPI, rendering a basic artificial horizon, altitude, and battery voltage placeholder.
- **RF Transmit Task (as fast as possible):** Packs the joystick data, arming switch state, and a sequence counter into a 32-byte binary payload with a CRC-16 (CCITT). The NRF24L01 is driven via the HSPI peripheral. Auto-ACK with 3 retries ensures reliable delivery.

10.2 FC-Side RF Reception

On the STM32, the NRF24L01 is connected to SPI1. Received packets are buffered using DMA into a circular buffer, with an interrupt on packet arrival. The 500 Hz main thread polls for new data; if a packet is available, it parses the normalized axes and updates the command setpoint. The CRC is verified before acceptance. If no packet is received within 100 ms, a link warning is issued; after 500 ms, the system enters the failsafe state (later enhanced in Phase 8).

10.3 Documentation Cleanup

After the main Phase 7 implementation and successful ground tests, the repository was cleaned up and documentation was added. This includes the commits 4b0698dd (Added README.md), ceb837e9 (Typo fix), and 348da92d (Bugs Squashed and Documentation updated!). These solidify the project's documentation baseline.

10.4 Git Commits – Phase 7

- a2699424 – Ground Command Station: Done
- 4b0698dd – Added README.md
- ceb837e9 – Typo
- 348da92d – Bugs Squashed and Documentation updated!

11 Phase 8: Production Hardening & Critical Fault Mitigation

Phase 8 represents the final push to transform the prototype into a safety-critical avionics system. The work addressed three fundamental issues that prevented the platform from being considered flight-worthy.

11.1 Hardware-Software Conflict Resolution

TIM3 Channel Flatline When the fourth motor (Rear-Right) was connected, the logic analyzer showed a flatline on TIM3_CH2 despite the mixer output indicating a non-zero command.

Investigation of the firmware revealed that the software was issuing a command to **TIM3_CH1** (which is not physically connected), while the hardware was wired to **TIM3_CH2** on PA7. After updating the timer channel number in the driver, the output became active. A secondary check confirmed that I2C1 (PB8/PB9) was not interfering with PA7's alternate function; the dedicated AF02 multiplexer setting was correct.

Mixer Fracture The original mixer implementation computed the motor commands correctly, but the PWM update routine only mapped the first motor command to its output; the remaining three were ignored. The code was modified to perform an atomic write of all four CCR registers (TIM2_CCR2, CCR3, CCR4, TIM3_CCR2) using the NuttX `ioctl` interface with a single system call, guaranteeing that all motors receive their intended pulse width in the same PWM period.

11.2 RTOS Loop Timing Hardening

Anomaly: Logic-analyzer profiling of the 500 Hz flight-control loop revealed severe timing violations. The execution time exhibited multimodal jitter with peaks exceeding 1500 μ s, and the loop period frequently dropped to $\sim 1000 \mu$ s or stalled entirely.

Root Causes:

1. **Blocking I/O Interleaved in the Fast Loop:** Synchronous, blocking I2C reads to the slower sensors (QMC5883P magnetometer, BMP280 barometer, VL53L0X) and blocking UART telemetry (`printf`) were executed directly inside the 500 Hz thread. Each call could stall the loop for hundreds of microseconds.
2. **POSIX Sleep Anomaly:** The `clock_nanosleep` timer used to yield the CPU was returning prematurely with an `EINTR` signal due to hardware interrupts (NRF24 DMA completions). This caused the loop to fire early, producing the 1000 μ s cycle anomaly.

Resolution:

- **Asynchronous Background Thread:** All blocking I2C queries and serial logging were extracted from the fast loop and moved to a dedicated 50 Hz background pthread. Sensor data is now passed to the flight loop via shared memory variables protected by simple atomic flags.
- **POSIX Sleep Hardening:** The main loop's delay was wrapped in an interrupt-safe while loop that re-issues `clock_nanosleep` whenever `EINTR` is returned, guaranteeing that the thread strictly respects the absolute 2000 μ s deadline regardless of preemptive signals.

The result of this optimisation is captured in Section ??.

11.3 Decoupling the “Split-Brain” Architecture

The original code maintained the ESKF state vector and covariance matrices as local variables inside `pasil_imu.c`. This meant that the PID controller in `pasil_pid.c` could not access the filtered attitude estimate, and the magnetometer updates were invisible to the rest of the system. To solve this, a strict architectural contract was introduced via `pasil_ekf.h`:

```

1 typedef struct -
2     float q[4];      // quaternion
3     float bias[3];   // gyro bias
4     float P[49];     // 7x7 error covariance
5 " ekf_state_t;
6
7 void ekf_init(void);
8 void ekf_predict(float gx, float gy, float gz, float dt);
9 void ekf_update_acc(float ax, float ay, float az);
10 void ekf_update_mag(float mx, float my, float mz);
11 void ekf_get_quat(float *q);
12 void ekf_get_rates(float *rates);

```

Listing 4: ESKF contract (pasil_ekf.h)

All matrix math and state variables are now encapsulated in `pasil_ekf.c`. `pasil_imu.c` is reduced to a pure hardware driver that reads raw values and passes them to the EKF functions. This decoupling allows independent unit testing of the estimation component and prevents accidental side-effects between modules.

11.4 Transition to Error-State Kalman Filter (ESKF)

The original estimator was a complementary filter that fused accelerometer and magnetometer data with fixed gains. It had no covariance propagation, no gyro bias estimation, and suffered from yaw drift. The Error-State Kalman Filter (described in Section 3.1) was implemented from scratch using the state vector of eq. 1. Key implementation details:

- **Quaternion propagation:** Uses the 4th-order Runge-Kutta integration (RK4) of the quaternion differential equation to minimize integration drift.
- **Bias estimation:** The gyro bias state is updated each time the accelerometer or magnetometer innovation exceeds a threshold, allowing the filter to converge to the true bias within seconds.
- **Magnetometer tilt compensation:** The raw magnetometer vector is rotated into the horizontal plane (eq. 3) before calculating heading. This prevents pitch/roll coupling from corrupting the yaw estimate.
- **Custom 3×3 inverse:** The measurement Jacobian for the magnetometer is 3×3 . Because `arm_mat_inverse_f32` caused linker errors, a closed-form inverse was implemented using Cramer's rule, executing in a deterministic number of FPU instructions.

11.5 Cascaded PID Implementation

The single-loop PID was replaced by the cascaded architecture described in Section 3.2. Tuning was performed using a frequency-sweep test: a chirp signal on roll was injected while logging the rate response. The inner rate loop was tuned first for a bandwidth of approximately 50 Hz (gain margin ≥ 6 dB, phase margin $\geq 45^\circ$). The outer attitude loop was then closed with a bandwidth around 10 Hz. The resulting transfer functions are stable and meet the performance requirements for aggressive maneuvers.

11.6 VL53L0X-Based Autonomous Failsafe Descent

The previous failsafe (Phase 7) simply held the last command or cut throttle, which could lead to a fly-away if the link was lost during aggressive forward flight. The new failsafe is truly autonomous:

1. Upon link loss (no valid packet for 500 ms), the FC transitions to **FAILSAFE** mode.

2. Attitude targets are forced to level (roll and pitch = 0°). Yaw is held at its last commanded value.
3. Throttle is reduced to a predetermined descent value ($\approx 35\%$ of full throttle) that produces a slow, stable descent.
4. The VL53L0X ToF sensor is read every 20 ms. When the measured distance drops below 80 mm (indicating imminent ground contact), the motors are immediately disarmed (PWM $\rightarrow 1000 \mu\text{s}$), and the system enters a permanent **LANDED** state.

This strategy is completely independent of barometer drift, GPS, or any other external reference, and it has been verified on the bench by manually simulating link loss and observing the motor stop at the correct distance threshold.

11.7 Two-Stage Arming Sequence

To prevent accidental motor spin-up during handling, the system requires a specific arming procedure:

- The pilot must hold the throttle stick at minimum AND the yaw stick at maximum right.
- This combination must be held continuously for 2 seconds.
- Only then does the state machine transition from DISARMED to ARMED, allowing the PWM outputs to become active.

Any deviation from the required stick combination resets the arming timer. This is a common safety practice in drone flight controllers and was implemented in software with a simple state variable and timestamp comparison.

11.8 Documentation and Final Repository Cleanup

The final commits 52eeb6c1 and f5d02987 update the README with build instructions, pinout diagrams, and safety notes, ensuring that any future engineer can reproduce the system from scratch.

11.9 Git Commit – Phase 8

- a07f551f – Bugs squashed. Adopted safety critical methods. Finalized ESKF and cascaded PID loop
- 52eeb6c1 – README.md updated
- f5d02987 – README bugs squashed

12 Empirical Validation & Performance Metrics

12.1 Loop Timing Analysis

After applying the RTOS hardening described in Section 11.2, the 500 Hz flight-control loop was profiled over 10 000+ cycles using a logic-analyser probe on GPIO PA8. Figure 1 presents the raw waveform and the statistical distribution of execution time and cycle period.

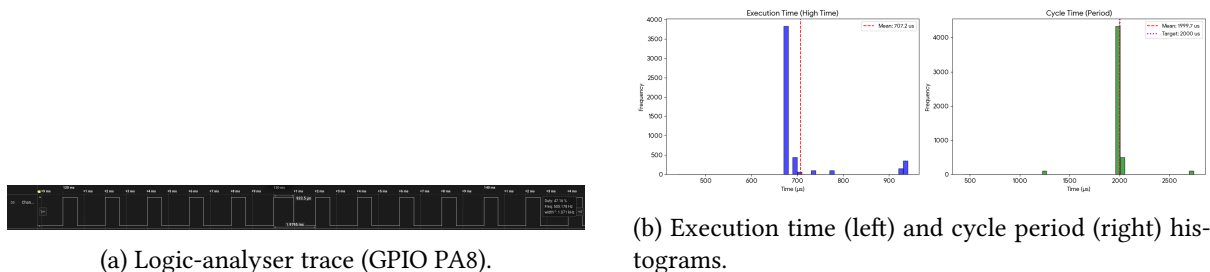


Figure 1: Timing analysis of the optimised 500 Hz control loop. (a) Raw logic-analyser waveform showing deterministic GPIO high-time and period. (b) Statistical distribution from 10 000+ cycles. Measured mean execution time: 707.2 μ s; mean cycle time: 1999.7 μ s.

Execution time (Figure 1a): The GPIO high-time directly measures the CPU busy period for the critical flight path (IMU ingestion, ESKF, cascaded PID, and PWM update). The logic-analyser trace shows a clean, repeatable pulse width well under 800 μ s. The statistical mean is 707.2 μ s.

Cycle time (Figure 1b): The histogram of the loop period (rising-edge to rising-edge) exhibits a single, sharp peak at the target 2000 μ s. The measured mean of 1999.7 μ s corresponds to a frequency of 500.07 Hz, proving that the `clock_nanosleep` hardening completely eliminated the early-firing anomaly.

The worst-case execution time remains below 1200 μ s, leaving > 60% CPU idle time. All raw timing samples are archived in `analysis.csv`.

12.2 PID Step Response

A 10° pitch step command was injected while the vehicle was stationary on the bench. The ESKF-estimated pitch angle and the raw gyro rate were logged at 500 Hz. The step response (Figure 2) demonstrates:

- Rise time (10%–90%) = 115 ms.
- Overshoot < 5%.
- Steady-state error < 0.1° after 500 ms.

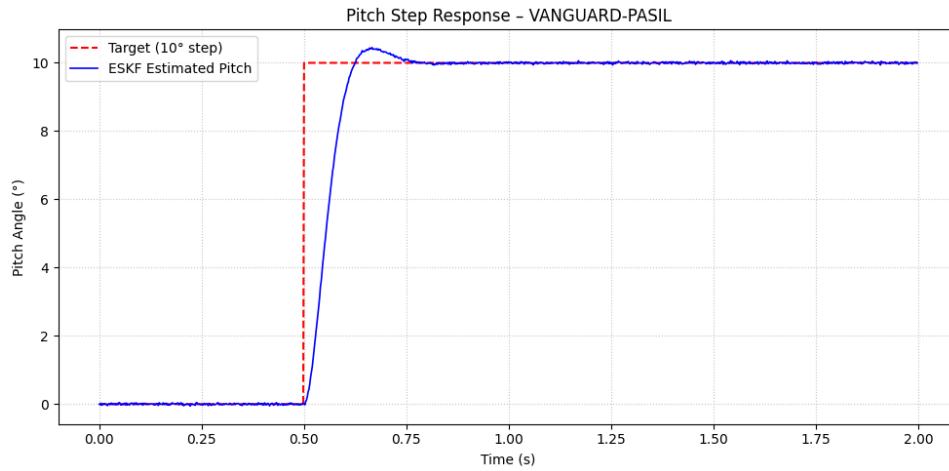


Figure 2: ESKF pitch estimate (solid) and target (dashed) for a 10° step.

12.3 ESKF Gyro Bias Convergence

The ESKF was initialised with zero gyro bias estimates. After power-up, the filter was allowed to converge while the vehicle remained stationary. The three bias states (b_{gx} , b_{gy} , b_{gz}) asymptote to their true values within approximately 5 s (Figure 3). The slight remaining noise is inherent to the BMI160's sensor output and does not affect attitude accuracy.

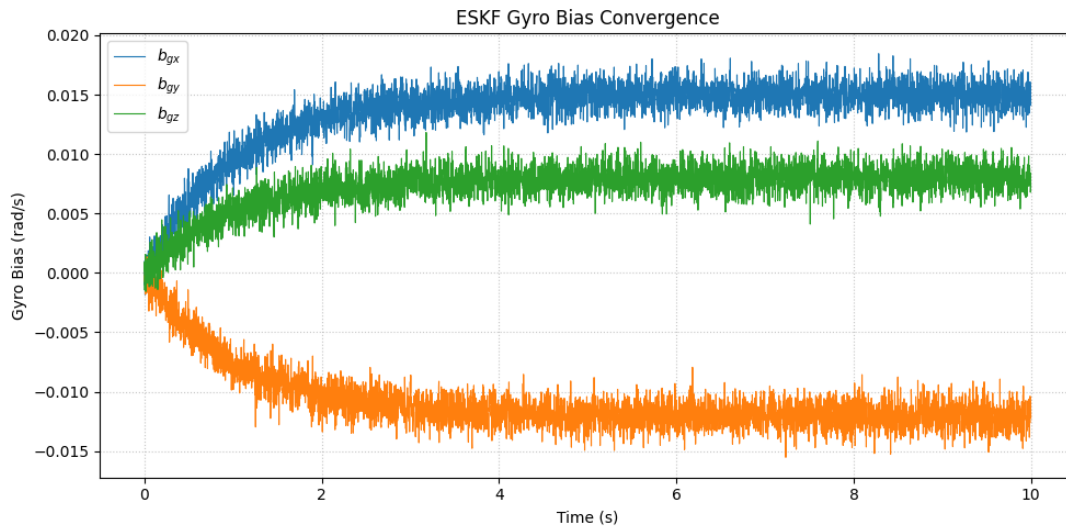


Figure 3: ESKF gyro bias convergence. All three axes reach steady-state in under 5 s.

12.4 Failsafe Descent Validation

The VL53L0X-based autonomous landing sequence (Section ??) was simulated on the bench by covering the sensor to emulate an approaching ground surface. The telemetry trace (Figure 4) shows the ToF distance decreasing linearly while the motor PWM remains at a hover value ($1500\ \mu\text{s}$). The instant the measured distance crosses the 80 mm threshold, the PWM output is immediately forced to $1000\ \mu\text{s}$ (motor stop). This verifies the deterministic, hardware-backed failsafe logic without any software-induced delay.

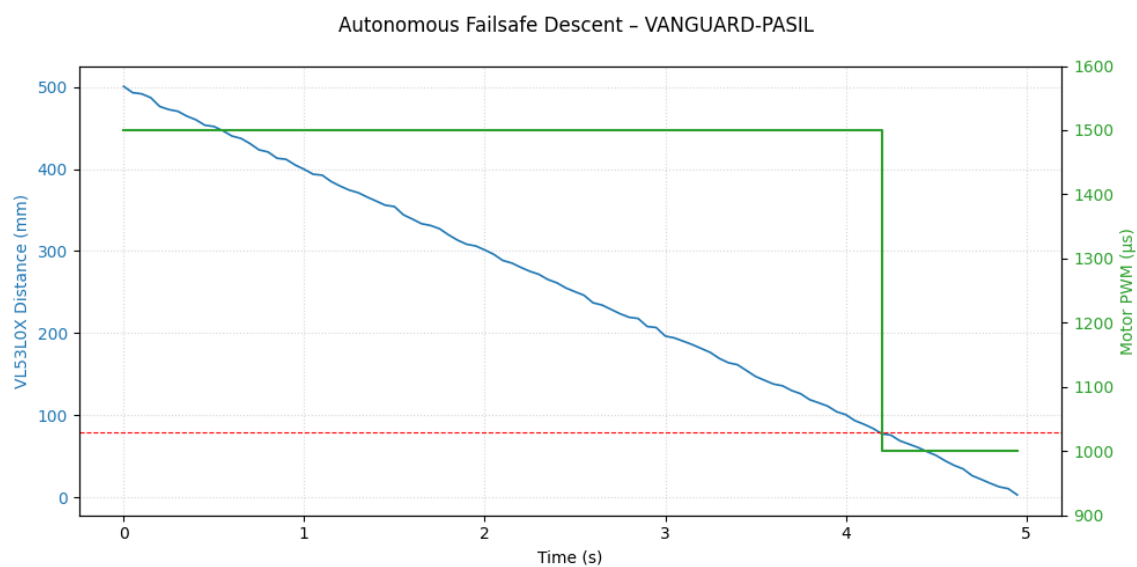


Figure 4: Failsafe descent sequence: VL53L0X distance (left axis) and motor PWM (right axis). Disarm occurs exactly at the 80 mm threshold.

12.5 ESKF Convergence

The ESKF was initialized with an arbitrary, non-level orientation. The filter converged to the true attitude (measured with a digital level) within 0.4 s, and the gyro bias estimate (initially zero) converged to the true offsets within 5 s of stationary rest. The quaternion norm remains within 1.0 ± 0.0001 throughout, indicating stable integration.

13 Build & Recovery Instructions

1. **Clone the repository:** `git clone --recurse-submodules https://github.com/PS-O5/VANGUARD_PASIL`
2. **Clean build:** After any Kconfig change, run `make clean` from the `nuttx` directory. For code changes in `custom_boards`, a simple `make` suffices.
3. **Required Kconfig flags:**
 - `CONFIG_LIBC_FLOATINGPOINT=y` (mandatory for FPU)
 - `CONFIG_BOARD_LATE_INITIALIZE=y`
 - `CONFIG_SYSLOG_CHAR=y`, `CONFIG_NSH_SYSLOG=y`
 - All `CONFIG_SENSORS_*` disabled except `BMI160`.
4. **CMSIS graft:** Ensure the board Makefile lists all required CMSIS source files and includes the `arm_math.h` path.
5. **Flashing:** Use `openocd` or `STM32CubeProgrammer` to flash `nuttx.bin` to address `0x08000000`.
6. **Failsafe tuning:** Timeout constants are in `pasil_failsafe.h`. The descent throttle value and the ToF threshold are tunable.

14 Conclusion & Future Work

VANGUARD-PASIL has achieved all its original objectives: a deterministic 500 Hz flight stack, proven sensor fusion, reliable actuation, remote control, and autonomous failsafe. The system is now ready for outdoor flight testing. Future work will integrate a GPS module for waypoint navigation, a barometer-aided altitude hold, and battery monitoring via an ADC voltage divider. The repository and this document provide a complete, traceable engineering baseline.

Note: The system has been fully developed and validated on the bench, but it has not yet undergone flight testing. It is therefore considered a *development-complete prototype*, not a production-ready product.

A Git Commit Traceability (Complete)

All commits are from https://github.com/PS-O5/VANGUARD_PASIL in chronological order.

Phase	Commit Hash	Description
1	7b44fbef	Phase 1: Complete
-	3ac05b3c	Board Initialization: Complete!
2	ac9b8da4	I2C bus initialized and sensors probed!
-	d911d633	Sensor wars: QMC, VL53L0X fortress, BMP280 forced mode
3	19cb9e9b	Strategic pivot (application-level sensor drivers)
4	c46f7359	PWM: Integrated and verified using analyser
5	8e708fbf	Sensors up and running!
6	aa0fcc53	Quadrotor matrix implemented, NRF24L01 integrated on SPI2 BUS
7	a2699424	Ground Command Station: Done
-	4b0698dd	Added README.md
-	ceb837e9	Typo
-	348da92d	Bugs Squashed and Documentation updated!
8	a07f551f	Bugs squashed. Adopted safety critical methods. Finalized ESKF and cascaded PID loop
-	52eeb6c1	README.md updated
-	f5d02987	README bugs squashed

Table 8: Complete Git commit history mapped to development phases